

Julien Brenneck

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EDUCATION	University of Massachusetts Amherst	
	PhD in Mathematics (Dropped out)	Sep 2021 – Oct 2022
	M.S. in Applied Mathematics	May 2021
	B.S. in Mathematics and Computer Science (Double Major)	Jan 2019
EXPERIENCE	NeuralFrame (Medical Data Startup)	
	Senior Engineer	May 2024 – Present
	• Lead backend development (2-5 people), growing market share from two to ten+ percent. • Built SOC2 secure data solutions for dozens of hospitals, including the VA on GovCloud. • Oversaw MongoDB to PostgreSQL migration, including a real-time data sync.	
	Software Engineer	Nov 2022 – May 2024
	• Rebuilt all test infrastructure, leading to large decrease in support costs. • Built and maintained modern web backends in FastAPI, unifying legacy Flask microservices.	
	UMass Amherst, Nanoelectronics Theory and Simulation Laboratory	
	Research Assistant	May 2017 – Oct 2022
	• Developed FEAST-type contour based eigensolver algorithms (included in Intel MKL). • Independently developed novel nonlinear eigenvalue algorithm outperforming previous techniques. • Worked within Electrical Engineering Dept, applications to semiconductor design.	
	UC Berkeley / Lawrence Berkeley National Laboratory, Materials Science	
	Research Assistant (SULI Program)	May 2018 – Sep 2018
	• Conducted literature review on machine learning methodology in materials informatics. • Designed experiments with materials data to highlight methodological breakdown of ML models. • Worked on job optimization in a compute pipeline/framework for supercomputer workflows in Python.	
SKILLS	Python, Julia, Haskell, JS/TS, Nix/NixOS, FORTRAN, Rust, Java, $\LaTeX$, SQL Numerical Algorithms, High Performance Compute, Parallel & Concurrent Programming	
PUBLICATIONS	<i>An Iterative Method for Contour-Based Nonlinear Eigensolvers</i> [arXiv]	
	• First author — a highly parallel iterative algorithm, all code is open source (Fortran/Julia).	
	<i>Rocketsled: A Software Library for Optimizing High-Throughput Computational Searches</i>	2019
	• Second Author — a modern compute pipeline enabling ML optimization for supercomputer workflows.	
POSTERS/TALKS	<i>Cross-Validation Methodology in Materials Science</i> [pdf]	2018
	<i>Bayesian Hyper-Parameter Optimization for Neural Networks</i> [pdf]	2018
	<i>Iteration for Contour-Based Nonlinear Eigensolvers</i> [pdf]	2020
PROJECTS	<i>GPT2.hs</i> : Large Lambda Model - NanoGPT Forward Pass in Haskell (hmatrix) [github]	2025
	<i>happrox</i> : A collection of approximation theory routines in Haskell [github]	2025
AWARDS	Outstanding Undergraduate Course Assistant	2017
	• For work as TA for CompSci 121: Introduction to Programming at UMass Amherst (awarded twice).	
	Scholastic Art and Writing National Gold Medal	2014
	• For work in generative computer graphics.	